

CSL253 - Theory of Computation

Tutorial 1

Team Members

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Question 4

Each of the following languages is the intersection of two simpler languages. In each part, construct DFAs for the simpler languages, then combine them using the construction to give the state diagram of a DFA for the language given. In all parts $Z = a, b$.

1. $\{w \mid w \text{ has at least three } a\text{'s and at least two } b\text{'s}\}$
2. $\{w \mid w \text{ has exactly two } a\text{'s and at least two } b\text{'s}\}$
3. $\{w \mid w \text{ has an even number of } a\text{'s and one or two } b\text{'s}\}$
4. $\{w \mid w \text{ has an even number of } a\text{'s and each } a \text{ is followed by at least one } b\}$
5. $\{w \mid w \text{ has an odd number of } a\text{'s and ends with a } b\}$
6. $\{w \mid w \text{ has even length and an odd number of } a\text{'s}\}$

a.Problem Description

Given an alphabet $\Sigma = \{a, b\}$, construct a DFA that recognizes strings that $\{w \mid w \text{ has at least three } a\text{'s and at least two } b\text{'s}\}$.

Solution Strategy

To solve this problem, we will follow these steps:

1. First, we construct a DFA M_1 that accepts strings containing at least three occurrences of 'a'
2. Next, we will construct DFA M_2 that accepts strings containing at least two occurrences of 'b'
3. Then we will merge the two DFA's M_1 and M_2 to get the DFA M
4. To achieve this, we define DFA states q_{xy} , where x tracks the count of 'a' (capped at 3) and y tracks the count of 'b' (capped at 2).
5. The DFA transitions update x when an 'a' is encountered and y when a 'b' is encountered, ensuring that once we reach three 'a's and two 'b's, we remain in the accepting state.
6. Finally, we define the initial state q_{00} (no 'a' or 'b' seen yet) and the accepting state q_{32} (at least three 'a's and two 'b's encountered).

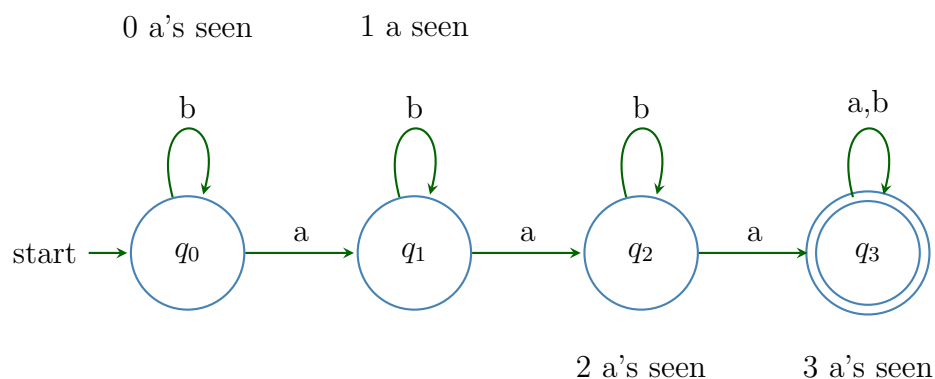
Solution

First we will write Regular Language for the strings having atleast 3 a's

Regular Language

$$A_1 = \Sigma^* a \Sigma^* a \Sigma^* a \Sigma^* = \{\omega \mid \omega \text{ is a string which has atleast 3 a's}\}$$

DFA Diagram



Defining the DFA

We define the DFA as a five-tuple $M_1 = (Q_1, \Sigma, \delta_1, q_0, F_1)$ where:

- **States:** $Q_1 = \{q_0, q_1, q_2\}$, where:
 - q_0 : Initial state where the input has 0 a's.
 - q_1 : State where the input has 1 a.
 - q_2 : State where the input has 2 a's.
 - q_3 : State where the input has atleast 3 a's.
- **Alphabet:** $\Sigma = \{a, b\}$
- **Transition Function:** $\delta_1 : Q_1 \times \Sigma \rightarrow Q_1$

$$\delta(\text{current state}, \text{input}) = \text{next state}$$

The transitions are defined as follows:

$$\begin{aligned}\delta_1(q_0, a) &= q_1 \\ \delta_1(q_0, b) &= q_0 \\ \delta_1(q_1, a) &= q_2 \\ \delta_1(q_1, b) &= q_1 \\ \delta_1(q_2, a) &= q_3 \\ \delta_1(q_2, b) &= q_2 \\ \delta_1(q_3, a) &= q_3 \\ \delta_1(q_3, b) &= q_3\end{aligned}$$

- **Initial State:** q_0
- **Acceptance States:** $F_1 = \{q_3\}$

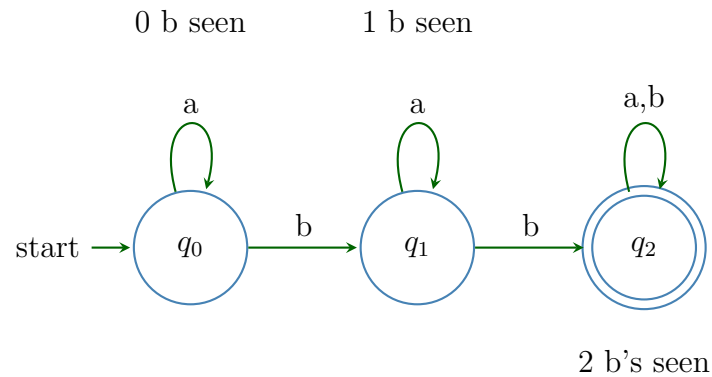
Transition Table

Current State	Input a	Input b
q_0	q_1	q_0
q_1	q_2	q_1
q_2	q_3	q_2
q_3	q_3	q_3

Now, we write the Regular Language for the strings having at least two b's.

Regular Language

$$A_2 = \Sigma^* b \Sigma^* b \Sigma^* = \{\omega \mid \omega \text{ is a string with at least two b's}\}$$



Defining the DFA

We define the DFA as a five-tuple $M_2 = (Q_2, \Sigma, \delta_2, q_0, F_2)$ where:

- **States:** $Q_2 = \{q_0, q_1, q_2\}$, where:
 - q_0 : Initial state where the input has 0 b's.
 - q_1 : State where the input has 1 b.
 - q_2 : State where the input has at least 2 b's.
- **Alphabet:** $\Sigma = \{a, b\}$
- **Transition Function:** $\delta_2 : Q_2 \times \Sigma \rightarrow Q_2$

$$\delta(\text{current state}, \text{input}) = \text{next state}$$

The transitions are defined as follows:

$$\begin{aligned}\delta_2(q_0, a) &= q_0 \\ \delta_2(q_0, b) &= q_1 \\ \delta_2(q_1, a) &= q_1 \\ \delta_2(q_1, b) &= q_2 \\ \delta_2(q_2, a) &= q_2 \\ \delta_2(q_2, b) &= q_2\end{aligned}$$

- **Initial State:** q_0
- **Acceptance States:** $F_2 = \{q_2\}$

Transition Table

Current State	Input a	Input b
q_0	q_0	q_1
q_1	q_1	q_2
q_2	q_2	q_2

Merging the DFA for atleast 3 a's and at least 3 b's

Now, we merge the two DFAs M_1 and M_2 by performing the following steps:

1. Create a new set of states by taking the Cartesian product of the states of the two DFAs. The new states are named as $q_i p_j$, where q_i is a state from the DFA for "atleast 3 a's" and p_j is a state from the DFA for "at least two b's".
2. Define the transitions for the merged DFA based on the transitions of the individual DFAs.
3. Mark a state as accepting if both corresponding states in the original DFAs are accepting.
4. Remove unreachable states from the merged DFA.

Defining the Merged DFA

We define the DFA as a five-tuple $M = (Q, \Sigma, \delta, q_{00}, F)$ where:

- **States:** $Q = \{q_{xy} \mid 0 \leq x \leq 3, 0 \leq y \leq 2\}$, where:
 - q_{xy} represents a state where:
 - * x is the number of a's encountered (capped at 3).
 - * y is the number of b's encountered (capped at 2).

- q_{00} is the initial state (no a's and no b's seen).
- q_{32} is the accepting state (at least 3 a's and at least 2 b's).

• **Alphabet:** $\Sigma = \{a, b\}$

• **Transition Function:** $\delta : Q \times \Sigma \rightarrow Q$, defined as:

$$\delta(q_{xy}, a) = q_{\min(x+1, 3), y}$$

$$\delta(q_{xy}, b) = q_{x, \min(y+1, 2)}$$

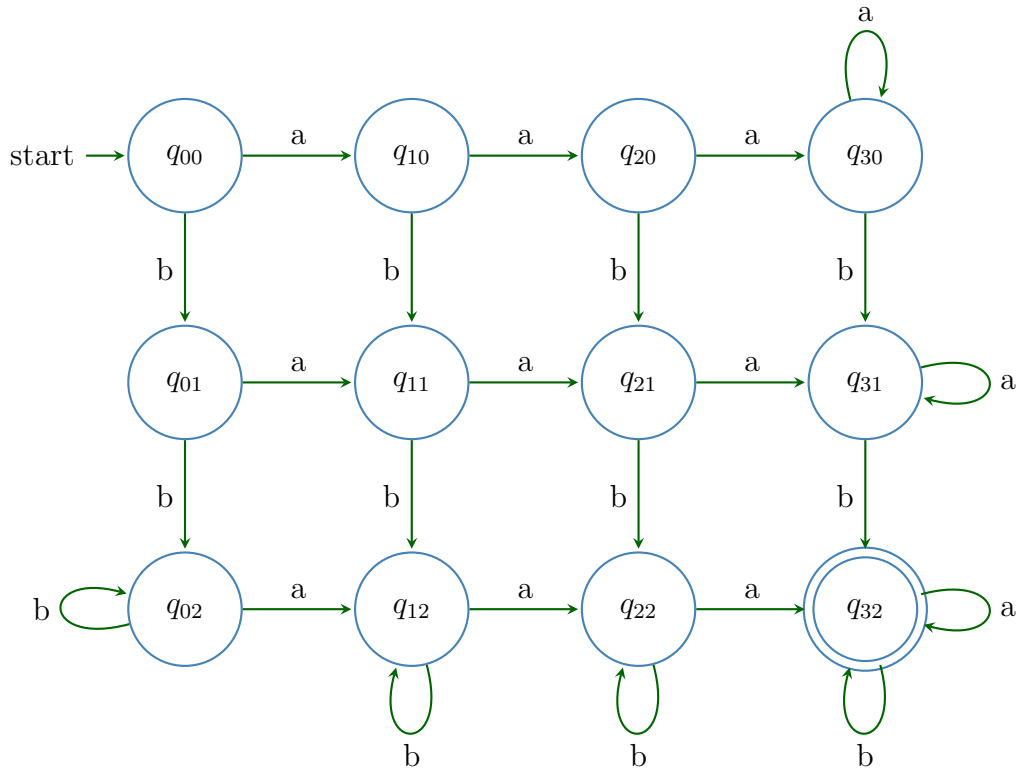
• **Initial State:** q_{00}

• **Acceptance State:** $F = \{q_{32}\}$

Transition Table for Merged DFA

Current State	Input a	Input b
q_{00}	q_{10}	q_{01}
q_{10}	q_{20}	q_{11}
q_{20}	q_{30}	q_{21}
q_{30}	q_{30}	q_{31}
q_{01}	q_{11}	q_{02}
q_{11}	q_{21}	q_{12}
q_{21}	q_{31}	q_{22}
q_{31}	q_{31}	q_{32}
q_{02}	q_{12}	q_{02}
q_{12}	q_{22}	q_{12}
q_{22}	q_{32}	q_{22}
q_{32}	q_{32}	q_{32}

Merged DFA Diagram



b.Problem Description

Given an alphabet $\Sigma = \{a, b\}$, construct a DFA that recognizes strings that $\{w \mid w \text{ has exactly two } a\text{'s and at least two } b\text{'s}\}$.

Solution Strategy

To solve this problem, we will follow these steps:

1. First, we construct a DFA M_1 that accepts strings containing exactly two occurrences of 'a'.
2. Next, we construct a DFA M_2 that accepts strings containing at least two occurrences of 'b'.
3. We then merge the two DFAs M_1 and M_2 using the Cartesian product method to create a combined DFA M that accepts strings satisfying both conditions.

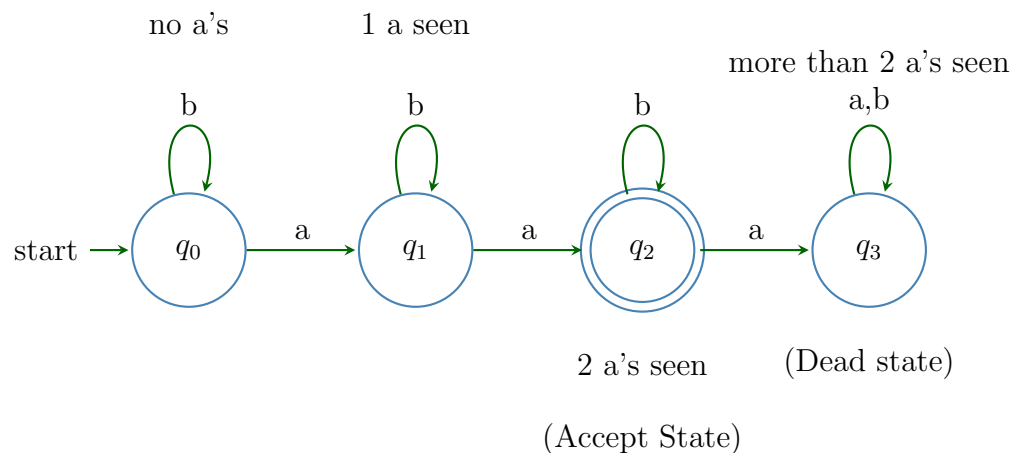
Solution

1. DFA for Exactly Two 'a's

Regular Language

$A_1 = b^*ab^*ab^* = \{w \mid w \text{ is a string with exactly two } a\text{'s}\}$

DFA Diagram



Defining the DFA

We define the DFA as a five-tuple $M_1 = (Q_1, \Sigma, \delta_1, q_0, F_1)$ where:

- **States:** $Q_1 = \{q_0, q_1, q_2, q_3\}$, where:
 - q_0 : Initial state (0 a 's seen).
 - q_1 : State after seeing 1 a .
 - q_2 : State after seeing exactly 2 a 's (accepting state).
 - q_3 : State after seeing more than 2 a 's (rejecting state).
- **Alphabet:** $\Sigma = \{a, b\}$.
- **Transition Function:** $\delta_1 : Q_1 \times \Sigma \rightarrow Q_1$, defined as:

$$\begin{aligned}\delta_1(q_0, a) &= q_1, & \delta_1(q_0, b) &= q_0, \\ \delta_1(q_1, a) &= q_2, & \delta_1(q_1, b) &= q_1, \\ \delta_1(q_2, a) &= q_3, & \delta_1(q_2, b) &= q_2, \\ \delta_1(q_3, a) &= q_3, & \delta_1(q_3, b) &= q_3.\end{aligned}$$

- **Initial State:** q_0 .
- **Acceptance States:** $F_1 = \{q_2\}$.

Transition Table

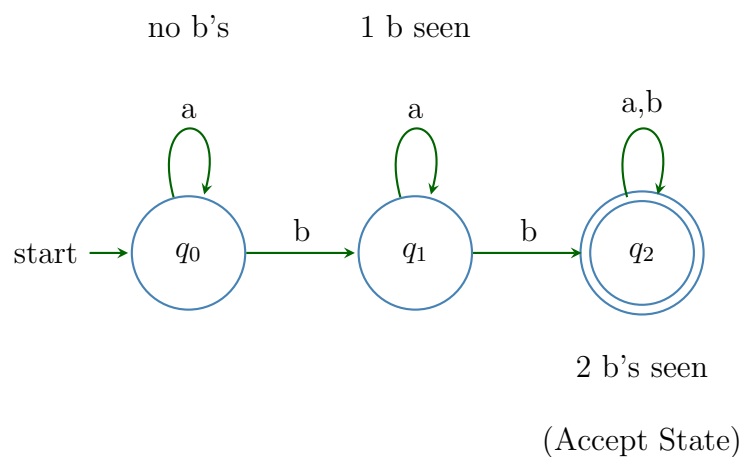
Current State	Input a	Input b
q_0	q_1	q_0
q_1	q_2	q_1
q_2	q_3	q_2
q_3	q_3	q_3

2. DFA for At Least Two 'b's

Regular Language

$$A_2 = \Sigma^* b \Sigma^* b \Sigma^* = \{w \mid w \text{ is a string with at least two } b\text{'s}\}$$

DFA Diagram



Defining the DFA

We define the DFA as a five-tuple $M_2 = (Q_2, \Sigma, \delta_2, q_0, F_2)$ where:

- **States:** $Q_2 = \{q_0, q_1, q_2\}$, where:
 - q_0 : Initial state (0 b 's seen).
 - q_1 : State after seeing 1 b .
 - q_2 : State after seeing at least 2 b 's (accepting state).
- **Alphabet:** $\Sigma = \{a, b\}$.
- **Transition Function:** $\delta_2 : Q_2 \times \Sigma \rightarrow Q_2$, defined as:

$$\begin{aligned}\delta_2(q_0, a) &= q_0, & \delta_2(q_0, b) &= q_1, \\ \delta_2(q_1, a) &= q_1, & \delta_2(q_1, b) &= q_2, \\ \delta_2(q_2, a) &= q_2, & \delta_2(q_2, b) &= q_2.\end{aligned}$$

- **Initial State:** q_0 .
- **Acceptance States:** $F_2 = \{q_2\}$.

Transition Table

Current State	Input a	Input b
q_0	q_0	q_1
q_1	q_1	q_2
q_2	q_2	q_2

3. Merged DFA for Exactly Two 'a's and At Least Two 'b's

Defining the Merged DFA

We define the merged DFA as a five-tuple $M = (Q, \Sigma, \delta, q_{00}, F)$ where:

- **States:** $Q = \{q_{xy} \mid 0 \leq x \leq 2, 0 \leq y \leq 2\}$, where:
 - x : Number of a 's seen .
 - y : Number of b 's seen
- **Alphabet:** $\Sigma = \{a, b\}$.
- **Transition Function:** $\delta : Q \times \Sigma \rightarrow Q$, defined as:

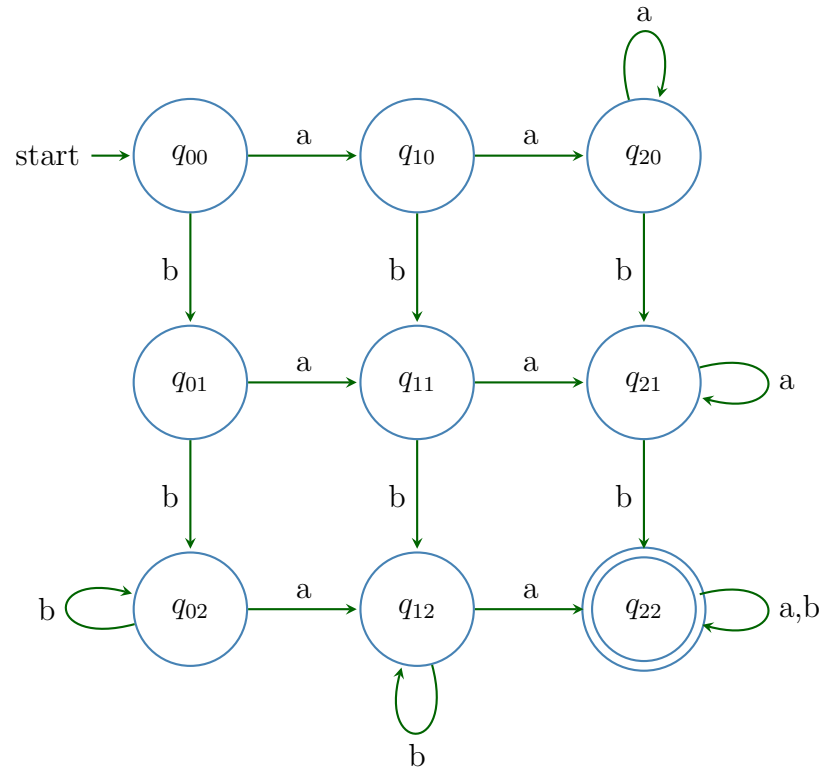
$$\begin{aligned}\delta(q_{xy}, a) &= q_{\min(x+1, 2), y}, \\ \delta(q_{xy}, b) &= q_{x, \min(y+1, 2)}.\end{aligned}$$

- **Initial State:** q_{00} .
- **Acceptance State:** $F = \{q_{22}\}$.

Transition Table for Merged DFA

Current State	Input a	Input b
q_{00}	q_{10}	q_{01}
q_{10}	q_{20}	q_{11}
q_{20}	q_{20}	q_{21}
q_{01}	q_{11}	q_{02}
q_{11}	q_{21}	q_{12}
q_{21}	q_{21}	q_{22}
q_{02}	q_{12}	q_{02}
q_{12}	q_{22}	q_{12}
q_{22}	q_{22}	q_{22}

Merged DFA Diagram



c.Problem Description

Given an alphabet $\Sigma = \{a, b\}$, construct a DFA that recognizes strings that $\{w \mid w \text{ has an even number of } a\text{'s and one or two } b\text{'s}\}$.

Solution Strategy

To solve this problem, we will follow these steps:

1. First, we construct a DFA M_1 that accepts strings containing an even number of a 's.
2. Next, we construct a DFA M_2 that accepts strings containing one or two b 's.
3. Finally, we combine M_1 and M_2 using the product construction to obtain a DFA M that recognizes strings with an even number of a 's and one or two b 's.

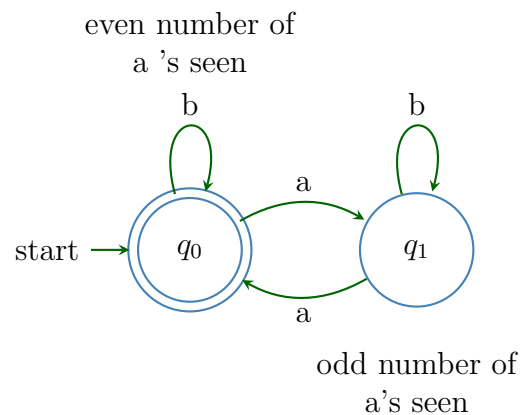
Solution

First, we construct the DFA M_1 for strings with an even number of a 's.

Regular Language for Even Number of a 's

$$A_1 = (b^*ab^*ab^*)^*$$

DFA Diagram for M_1



Defining the DFA M_1

We define the DFA as a five-tuple $M_1 = (Q_1, \Sigma, \delta_1, q_0, F_1)$ where:

- **States:** $Q_1 = \{q_0, q_1\}$, where:

- q_0 : State with an even number of a 's.
- q_1 : State with an odd number of a 's.
- **Alphabet:** $\Sigma = \{a, b\}$
- **Transition Function:** $\delta_1 : Q_1 \times \Sigma \rightarrow Q_1$

$$\delta(\text{current state}, \text{input}) = \text{next state}$$

The transitions are defined as follows:

$$\delta_1(q_0, a) = q_1$$

$$\delta_1(q_0, b) = q_0$$

$$\delta_1(q_1, a) = q_0$$

$$\delta_1(q_1, b) = q_1$$

- **Initial State:** q_0
- **Acceptance States:** $F_1 = \{q_0\}$

Transition Table for M_1

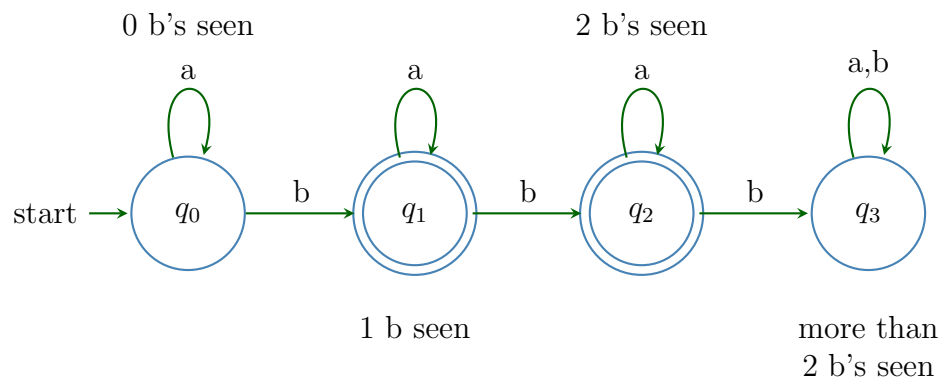
Current State	Input a	Input b
q_0	q_1	q_0
q_1	q_0	q_1

Next, we construct the DFA M_2 for strings with one or two b 's.

Regular Language for One or Two b 's

$$A_2 = (a^*ba^*) \cup (a^*ba^*ba^*)$$

DFA Diagram for M_2



Defining the DFA M_2

We define the DFA as a five-tuple $M_2 = (Q_2, \Sigma, \delta_2, q_0, F_2)$ where:

- **States:** $Q_2 = \{q_0, q_1, q_2, q_3\}$, where:
 - q_0 : State with 0 b 's.
 - q_1 : State with 1 b .
 - q_2 : State with 2 b 's.
 - q_3 : State with more than 2 b 's.
- **Alphabet:** $\Sigma = \{a, b\}$
- **Transition Function:** $\delta_2 : Q_2 \times \Sigma \rightarrow Q_2$

$$\delta(\text{current state}, \text{input}) = \text{next state}$$

The transitions are defined as follows:

$$\begin{aligned}\delta_2(q_0, a) &= q_0 \\ \delta_2(q_0, b) &= q_1 \\ \delta_2(q_1, a) &= q_1 \\ \delta_2(q_1, b) &= q_2 \\ \delta_2(q_2, a) &= q_2 \\ \delta_2(q_2, b) &= q_3 \\ \delta_2(q_3, a) &= q_3 \\ \delta_2(q_3, b) &= q_3\end{aligned}$$

- **Initial State:** q_0
- **Acceptance States:** $F_2 = \{q_1, q_2\}$

Transition Table for M_2

Current State	Input a	Input b
q_0	q_0	q_1
q_1	q_1	q_2
q_2	q_2	q_3
q_3	q_3	q_3

Merging the DFAs M_1 and M_2

Now, we merge the two DFAs M_1 and M_2 by performing the following steps:

1. Create a new set of states by taking the Cartesian product of the states of the two DFAs. The new states are named as $q_i p_j$, where q_i is a state from M_1 and p_j is a state from M_2 .
2. Define the transitions for the merged DFA based on the transitions of the individual DFAs.
3. Mark a state as accepting if both corresponding states in the original DFAs are accepting.
4. Remove unreachable states from the merged DFA.

Defining the Merged DFA M

We define the DFA as a five-tuple $M = (Q, \Sigma, \delta, q_{00}, F)$ where:

- **States:** $Q = \{q_i p_j \mid q_i \in Q_1, p_j \in Q_2\}$, where:
 - $q_i p_j$ represents a state where:
 - * q_i is the state of M_1 (even or odd number of a 's).
 - * p_j is the state of M_2 (number of b 's).
 - q_{00} is the initial state (even a 's and 0 b 's).
 - q_{01} and q_{02} are accepting states (even a 's and 1 or 2 b 's).
- **Alphabet:** $\Sigma = \{a, b\}$
- **Transition Function:** $\delta : Q \times \Sigma \rightarrow Q$, defined as:

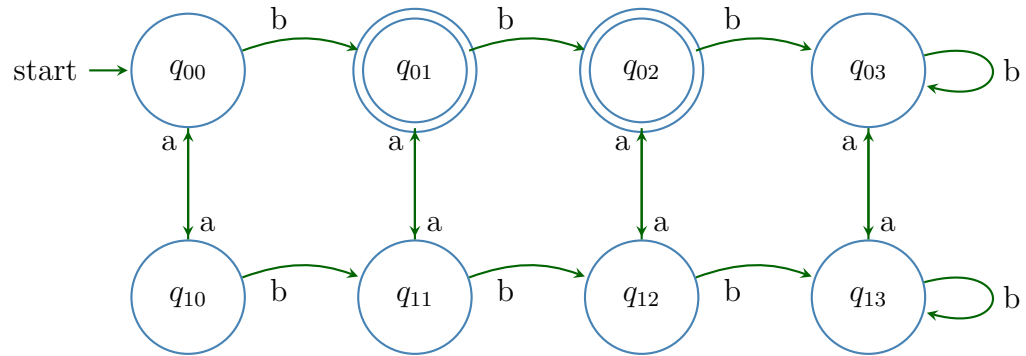
$$\delta(q_i p_j, a) = q_{\delta_1(q_i, a)} p_j \quad \text{and} \quad \delta(q_i p_j, b) = q_i p_{\delta_2(p_j, b)}$$

- **Initial State:** q_{00}
- **Acceptance States:** $F = \{q_{01}, q_{02}\}$

Transition Table for Merged DFA M

Current State	Input a	Input b
q_{00}	q_{10}	q_{01}
q_{10}	q_{00}	q_{11}
q_{01}	q_{11}	q_{02}
q_{11}	q_{01}	q_{12}
q_{02}	q_{12}	q_{03}
q_{12}	q_{02}	q_{13}
q_{03}	q_{13}	q_{03}
q_{13}	q_{03}	q_{13}

Merged DFA Diagram



d.Problem Description

Given an alphabet $\Sigma = \{a, b\}$, construct a DFA that recognizes strings that $\{w \mid w \text{ has an even number of } a\text{'s and each } a \text{ is followed by at least one } b\}$.

Solution Strategy

To solve this problem, we will follow these steps:

1. First, we construct a DFA M_1 that accepts strings containing an even number of 'a's.
2. Next, we construct a DFA M_2 that ensures each 'a' is followed by at least one 'b'.
3. Finally, we combine M_1 and M_2 using the Cartesian product method to create a DFA that recognizes strings with an even number of 'a's and each 'a' followed by at least one 'b'.

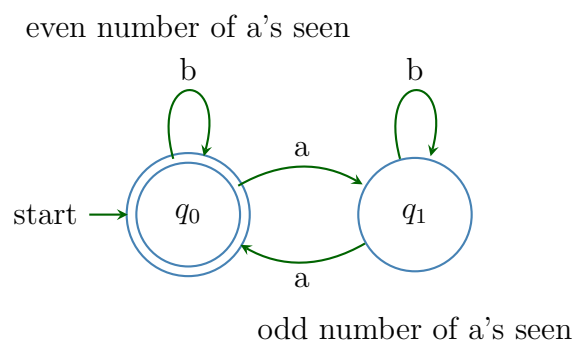
Solution

First we will write Regular Language for the strings having an even number of 'a's.

Regular Language

$$A_1 = (b^*ab^*ab^*)^* = \{\omega \mid \omega \text{ is a string with an even number of } a\text{'s}\}$$

DFA Diagram



Defining the DFA

We define the DFA as a five-tuple $M_1 = (Q_1, \Sigma, \delta_1, q_0, F_1)$ where:

- **States:** $Q_1 = \{q_0, q_1\}$, where:

- q_0 : Initial state where the input has an even number of 'a's and since 0 is also an even number so the start state will also be the accept state
- q_1 : State where the input has an odd number of 'a's.

• **Alphabet:** $\Sigma = \{a, b\}$

• **Transition Function:** $\delta_1 : Q_1 \times \Sigma \rightarrow Q_1$

$$\delta(\text{current state}, \text{input}) = \text{next state}$$

The transitions are defined as follows:

$$\delta_1(q_0, a) = q_1$$

$$\delta_1(q_0, b) = q_0$$

$$\delta_1(q_1, a) = q_0$$

$$\delta_1(q_1, b) = q_1$$

• **Initial State:** q_0

• **Acceptance States:** $F_1 = \{q_0\}$

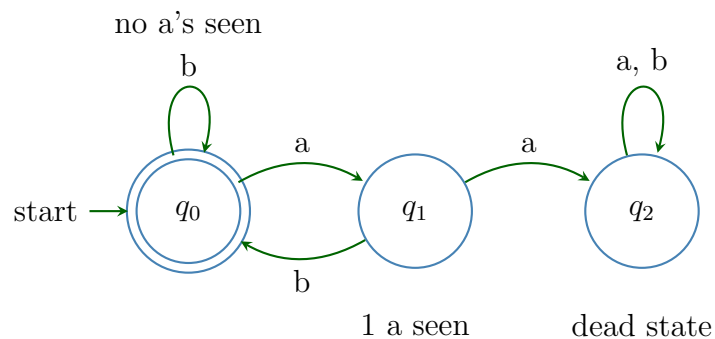
Transition Table

Current State	Input a	Input b
q_0	q_1	q_0
q_1	q_0	q_1

Now, we write the Regular Language for the strings where each 'a' is followed by at least one 'b'.

Regular Language

$$A_2 = (b^*ab^+)^*b^* = \{\omega \mid \omega \text{ is a string where each 'a' is followed by at least one 'b'}\}$$



Defining the DFA

We define the DFA as a five-tuple $M_2 = (Q_2, \Sigma, \delta_2, q_0, F_2)$ where:

- **States:** $Q_2 = \{q_0, q_1\}$, where:
 - q_0 : Initial state where no 'a' is immediately followed by another 'a'.
 - q_1 : State where an 'a' has been encountered and is waiting for a 'b'.
- **Alphabet:** $\Sigma = \{a, b\}$
- **Transition Function:** $\delta_2 : Q_2 \times \Sigma \rightarrow Q_2$

$$\delta(\text{current state}, \text{input}) = \text{next state}$$

The transitions are defined as follows:

$$\begin{aligned}\delta_2(q_0, a) &= q_1 \\ \delta_2(q_0, b) &= q_0 \\ \delta_2(q_1, a) &= q_1 \\ \delta_2(q_1, b) &= q_0\end{aligned}$$

- **Initial State:** q_0
- **Acceptance States:** $F_2 = \{q_0\}$

Transition Table

Current State	Input a	Input b
q_0	q_1	q_0
q_1	q_1	q_0

Merging the DFA for even number of 'a's and each 'a' followed by at least one 'b'

Now, we merge the two DFAs M_1 and M_2 by performing the following steps:

1. Create a new set of states by taking the Cartesian product of the states of the two DFAs. The new states are named as $q_i p_j$, where q_i is a state from the DFA for "even number of 'a's" and p_j is a state from the DFA for "each 'a' followed by at least one 'b'".
2. Define the transitions for the merged DFA based on the transitions of the individual DFAs.
3. Mark a state as accepting if both corresponding states in the original DFAs are accepting.
4. Remove unreachable states from the merged DFA.

Defining the Merged DFA

We define the DFA as a five-tuple $M = (Q, \Sigma, \delta, q_{00}, F)$ where:

- **States:** $Q = \{q_{00}, q_{01}, q_{10}, q_{11}\}$, where:
 - q_{00} : Initial state (even 'a's, no pending 'b').
 - q_{01} : State (even 'a's, waiting for 'b').
 - q_{10} : State (odd 'a's, no pending 'b').
 - q_{11} : State (odd 'a's, waiting for 'b').
- **Alphabet:** $\Sigma = \{a, b\}$
- **Transition Function:** $\delta : Q \times \Sigma \rightarrow Q$, defined as:

$$\delta(q_{00}, a) = q_{11}$$

$$\delta(q_{00}, b) = q_{00}$$

$$\delta(q_{01}, a) = q_d$$

$$\delta(q_{01}, b) = q_{01}$$

$$\delta(q_{10}, a) = q_{01}$$

$$\delta(q_{10}, b) = q_{10}$$

$$\delta(q_{11}, a) = q_d$$

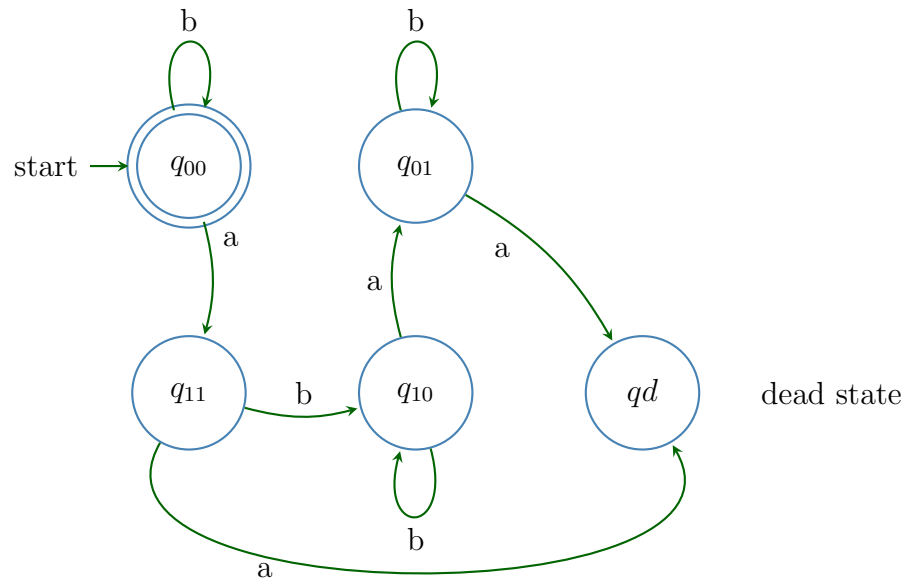
$$\delta(q_{11}, b) = q_{10}$$

- **Initial State:** q_{00}
- **Acceptance State:** $F = \{q_{00}\}$

Transition Table for Merged DFA

Current State	Input a	Input b
q_{00}	q_{11}	q_{00}
q_{01}	q_d	q_{01}
q_{10}	q_{01}	q_{10}
q_{11}	q_d	q_{10}

Merged DFA Diagram



e.Problem Description

Given an alphabet $\Sigma = \{a, b\}$, construct a DFA that recognizes strings w such that w has an odd number of a 's and ends with b .

Solution Strategy

To solve this problem, we will follow these steps:

1. Construct a DFA M_1 that accepts strings with an odd number of a 's.
2. Construct a DFA M_2 that accepts strings ending with a b .
3. Combine M_1 and M_2 using the product construction to create a DFA M that accepts strings satisfying both conditions.

Solution

DFA for Odd Number of a 's (M_1)

The DFA M_1 for strings with an odd number of a 's can be constructed as follows:

- States: $Q_1 = \{q_0, q_1\}$, where:
 - q_0 : Even number of a 's (initial state).
 - q_1 : Odd number of a 's (accepting state).
- Transitions:

$$\delta_1(q_0, a) = q_1$$

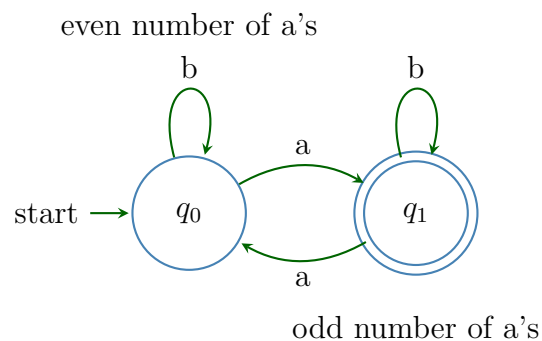
$$\delta_1(q_0, b) = q_0$$

$$\delta_1(q_1, a) = q_0$$

$$\delta_1(q_1, b) = q_1$$

- Acceptance: $F_1 = \{q_1\}$.

DFA Diagram for M_1



DFA for Strings Ending with b (M_2)

The DFA M_2 for strings ending with a b can be constructed as follows:

- States: $Q_2 = \{p_0, p_1\}$, where:
 - p_0 : Initial state (string does not end with b).
 - p_1 : Accepting state (string ends with b).
- Transitions:

$$\delta_2(p_0, a) = p_0$$

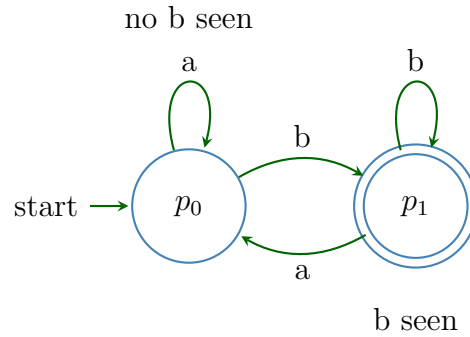
$$\delta_2(p_0, b) = p_1$$

$$\delta_2(p_1, a) = p_0$$

$$\delta_2(p_1, b) = p_1$$

- Acceptance: $F_2 = \{p_1\}$.

DFA Diagram for M_2



Merging M_1 and M_2 into M

The combined DFA M is defined as follows:

- States: $Q = Q_1 \times Q_2 = \{(q_0, p_0), (q_0, p_1), (q_1, p_0), (q_1, p_1)\}$.
- Transitions:

$$\delta((q_0, p_0), a) = (q_1, p_0)$$

$$\delta((q_0, p_0), b) = (q_0, p_1)$$

$$\delta((q_0, p_1), a) = (q_1, p_0)$$

$$\delta((q_0, p_1), b) = (q_0, p_1)$$

$$\delta((q_1, p_0), a) = (q_0, p_0)$$

$$\delta((q_1, p_0), b) = (q_1, p_1)$$

$$\delta((q_1, p_1), a) = (q_0, p_0)$$

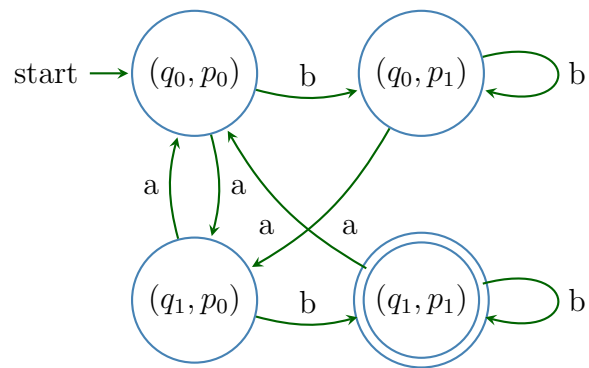
$$\delta((q_1, p_1), b) = (q_1, p_1)$$

- Initial State: (q_0, p_0) .
- Acceptance: $F = \{(q_1, p_1)\}$.

Transition Table for M

Current State	Input a	Input b
(q_0, p_0)	(q_1, p_0)	(q_0, p_1)
(q_0, p_1)	(q_1, p_1)	(q_0, p_1)
(q_1, p_0)	(q_0, p_0)	(q_1, p_1)
(q_1, p_1)	(q_0, p_0)	(q_1, p_1)

DFA Diagram for M



f.Problem Description

Given an alphabet $\Sigma = \{a, b\}$, construct a DFA that recognizes strings w such that w has even length and an odd number of a 's.

Solution Strategy

To solve this problem, we will follow these steps:

1. Construct a DFA M_1 that accepts strings of even length.
2. Construct a DFA M_2 that accepts strings with an odd number of a 's.
3. Combine M_1 and M_2 using the product construction to create a DFA M that accepts strings satisfying both conditions.

Solution

DFA for Even Length (M_1)

The DFA M_1 for strings of even length can be constructed as follows:

- States: $Q_1 = \{q_0, q_1\}$, where:
 - q_0 : Even length (initial and accepting state).
 - q_1 : Odd length.
- Transitions:

$$\delta_1(q_0, a) = q_1$$

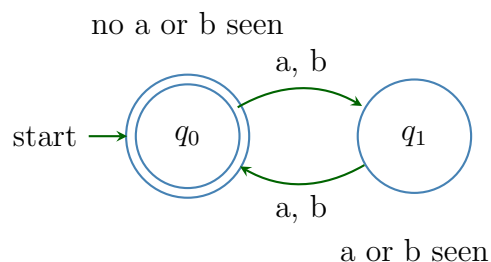
$$\delta_1(q_0, b) = q_1$$

$$\delta_1(q_1, a) = q_0$$

$$\delta_1(q_1, b) = q_0$$

- Acceptance: $F_1 = \{q_0\}$.

DFA Diagram for M_1



DFA for Odd Number of a 's (M_2)

The DFA M_2 for strings with an odd number of a 's can be constructed as follows:

- States: $Q_2 = \{p_0, p_1\}$, where:
 - p_0 : Even number of a 's (initial state).
 - p_1 : Odd number of a 's (accepting state).
- Transitions:

$$\delta_2(p_0, a) = p_1$$

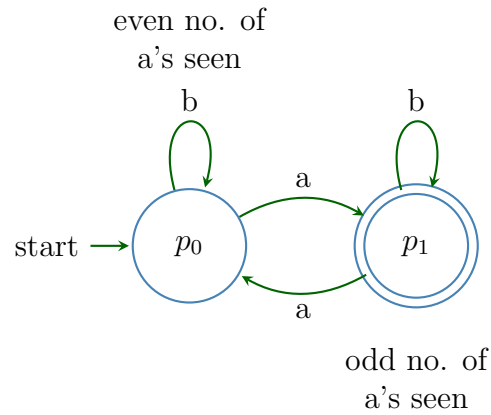
$$\delta_2(p_0, b) = p_0$$

$$\delta_2(p_1, a) = p_0$$

$$\delta_2(p_1, b) = p_1$$

- Acceptance: $F_2 = \{p_1\}$.

DFA Diagram for M_2



Merging M_1 and M_2 into M

The combined DFA M is defined as follows:

- States: $Q = Q_1 \times Q_2 = \{(q_0, p_0), (q_0, p_1), (q_1, p_0), (q_1, p_1)\}$.

- Transitions:

$$\delta((q_0, p_0), a) = (q_1, p_1)$$

$$\delta((q_0, p_0), b) = (q_1, p_0)$$

$$\delta((q_0, p_1), a) = (q_1, p_0)$$

$$\delta((q_0, p_1), b) = (q_1, p_1)$$

$$\delta((q_1, p_0), a) = (q_0, p_1)$$

$$\delta((q_1, p_0), b) = (q_0, p_0)$$

$$\delta((q_1, p_1), a) = (q_0, p_0)$$

$$\delta((q_1, p_1), b) = (q_0, p_1)$$

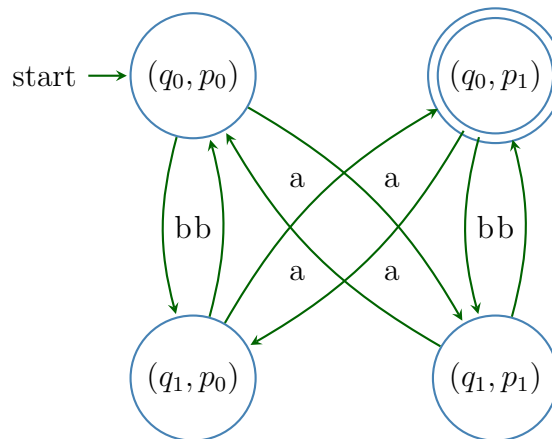
- Initial State: (q_0, p_0) .

- Acceptance: $F = \{(q_0, p_1)\}$.

Transition Table for M

Current State	Input a	Input b
(q_0, p_0)	(q_1, p_1)	(q_1, p_0)
(q_0, p_1)	(q_1, p_0)	(q_1, p_1)
(q_1, p_0)	(q_0, p_1)	(q_0, p_0)
(q_1, p_1)	(q_0, p_0)	(q_0, p_1)

DFA Diagram for M



Question 8

We want to construct an automaton over the alphabet $\{0, 1\}$, which recognizes words where:

- The number of ones at even-numbered positions is even.
- The number of ones at odd-numbered positions is odd.

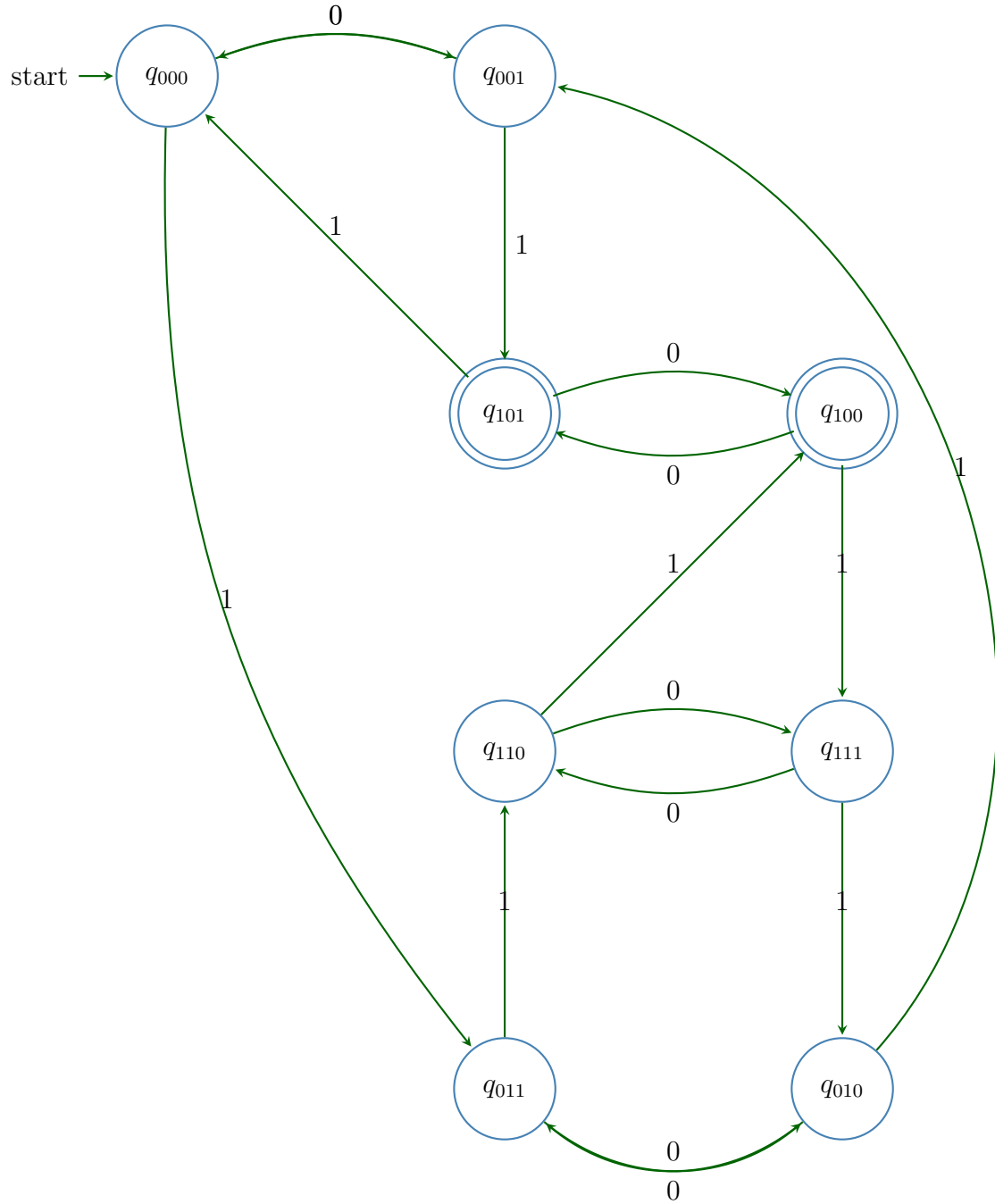
Solution

Automaton for the given language

DFA Diagram

Each state transitions based on the input symbol (0 or 1).

- When encountering a 0, it does not change the parity but remains in the current state.
- When encountering a 1, it toggles the parity of either the even or odd position, leading to a transition to a different state.



Defining the DFA

We define the DFA as a five-tuple $M = (Q, \Sigma, \delta, q_{001}, F)$ where:

- **States:** $Q = \{q_{000}, q_{001}, q_{100}, q_{101}, q_{110}, q_{111}, q_{010}, q_{011}\}$, q_{xyz} where x denotes x number of 1's at odd positions y denotes y number of 1's at even positions z denotes the parity of the input position
- **Alphabet:** $\Sigma = \{0, 1\}$

- **Transition Function:** $\delta : Q \times \Sigma \rightarrow Q$

$$\delta(\text{current state}, \text{input}) = \text{next state}$$

The transitions are defined as follows:

$$\begin{aligned} \delta(q_{000}, 0) &= q_{001} \\ \delta(q_{000}, 1) &= q_{011} \\ \delta(q_{001}, 0) &= q_{000} \\ \delta(q_{001}, 1) &= q_{100} \\ \delta(q_{100}, 0) &= q_{101} \\ \delta(q_{100}, 1) &= q_{111} \\ \delta(q_{101}, 0) &= q_{100} \\ \delta(q_{101}, 1) &= q_{000} \\ \delta(q_{110}, 0) &= q_{111} \\ \delta(q_{110}, 1) &= q_{101} \\ \delta(q_{111}, 0) &= q_{110} \\ \delta(q_{111}, 1) &= q_{010} \\ \delta(q_{010}, 0) &= q_{011} \\ \delta(q_{010}, 1) &= q_{001} \\ \delta(q_{011}, 0) &= q_{010} \\ \delta(q_{011}, 1) &= q_{110} \end{aligned}$$

- **Initial State:** q_{001}
- **Acceptance States:** $F = \{q_{101}, q_{100}\}$

Transition Table

Current State	Input 0	Input 1
q_{000}	q_{001}	q_{011}
q_{001}	q_{000}	q_{100}
q_{100}	q_{101}	q_{111}
q_{101}	q_{100}	q_{000}
q_{110}	q_{111}	q_{101}
q_{111}	q_{110}	q_{010}
q_{010}	q_{011}	q_{001}
q_{011}	q_{010}	q_{110}

““